

Span, Basis, and Dimension

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Matrix Multiplication: Context of Ramayan

A: FATHER	Dasharath	Kaushalya	Ram	Seeta	Janak
Dasharath	0	0	1	0	0
Kaushalya	0	0	0	0	0
Ram	0	0	0	0	0
Seeta	0	0	0	0	0
Janak	0	0	0	1	0

$$A = \begin{pmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix}$$

Matrix Multiplication: Context of Ramayan

B: WIFE	Dasharath	Kaushalya	Ram	Seeta	Janak
Dasharath	0	0	0	0	0
Kaushalya	1	0	0	0	0
Ram	0	0	0	0	0
Seeta	0	0	1	0	0
Janak	0	0	0	0	0

$$B = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Matrix Multiplication: Context of Ramayan

AB: WIFE'S FATHER	Dasharath	Kaushalya	Ram	Seeta	Janak
Dasharath	0	0	0	0	0
Kaushalya	0	0	0	0	0
Ram	0	0	0	0	0
Seeta	0	0	0	0	0
Janak	0	0	1	0	0

Matrix Multiplication: Context of Ramayan

BA: FATHER'S WIFE	Dasharath	Kaushalya	Ram	Seeta	Janak
Dasharath	0	0	0	0	0
Kaushalya	0	0	1	0	0
Ram	0	0	0	0	0
Seeta	0	0	0	0	0
Janak	0	0	0	0	0

Matrix Multiplication: F◦R◦I◦E◦N◦D◦S

Handwritten notes:
Alam — Malt
 Ila
Kian
Chanchal

F: Friends	Alam	Kian	Chanchal	Matt	Ila
Alam	0	0	1	1	1
Kian	0	0	0	1	1
Chanchal	1	0	0	1	0
Matt	1	1	1	0	0
Ila	1	1	0	0	0

Matrix Multiplication: $F \circ R \circ I \circ E \circ N \circ D \circ S$

F²: MUTUAL FRIENDS	Alam	Kian	Chanchal	Matt	Ila
Alam	3	2	1	1	0
Kian	2	2	1	0	0
Chanchal	1	1	2	1	1
Matt	1	0	1	3	2
Ila	0	0	1	2	2

Example: Span

Example (Span in $\mathbb{R}^3$)

Determine whether the vectors $\mathbf{v}_1 = (1, -1, 4)$, $\mathbf{v}_2 = (-2, 1, 3)$ and $\mathbf{v}_3 = (4, -3, 5)$ span $\mathbb{R}^3$.

Pick any point (vector in $\mathbb{R}^3$) as $(x_1, x_2, x_3) \in \mathbb{R}^3$.

$$\begin{aligned}(x_1, x_2, x_3) &= a \mathbf{v}_1 + b \mathbf{v}_2 + c \mathbf{v}_3 \\&= a(1, -1, 4) + b(-2, 1, 3) + c(4, -3, 5) \\&= (a - 2b + 4c, -a + b - 3c, 4a + 3b + 5c)\end{aligned}$$

$$\left. \begin{aligned}a - 2b + 4c &= x_1 \\-a + b - 3c &= x_2 \\4a + 3b + 5c &= x_3\end{aligned} \right\} \begin{array}{l} \text{by equating component-} \\ \text{wise.} \end{array}$$

Qn: Can we find $a, b, c \in \mathbb{R}$ for any $(x_1, x_2, x_3) \in \mathbb{R}^3$?

Example: Span

$$\dim(\mathcal{M}_2(\mathbb{R})) = 4.$$

Example (Span in $\mathcal{M}_2(\mathbb{R})$)

Verify that

$$A_1 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, A_2 = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}, A_3 = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}, \text{ and } A_4 = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$$

span $\mathcal{M}_2(\mathbb{R})$.

$$a, b, c, d \in \mathbb{R};$$

$$\begin{aligned} \underbrace{\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}}_{\substack{\in \mathcal{M}_{2 \times 2}(\mathbb{R}) \\ = \mathcal{M}_2(\mathbb{R})}} &= aA + bA_2 + cA_3 + dA_4 \\ &= a \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + b \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix} + c \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} + d \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \\ &= \begin{pmatrix} a+b+c+d & b+c+d \\ c+d & d \end{pmatrix} \end{aligned}$$

back substitution

$$\left. \begin{aligned} a+b+c+d &= a_{11} \\ b+c+d &= a_{12} \\ c+d &= a_{21} \\ d &= a_{22} \end{aligned} \right\}$$

Qn: Can we find $a, b, c, d \in \mathbb{R}$
for any arbitrary elements
 $\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \in M_2(\mathbb{R})$

ANS

$$\begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix} = \begin{pmatrix} a_{11} \\ a_{12} \\ a_{21} \\ a_{22} \end{pmatrix}$$

YES

Qn: Is this set can be a candidate for basis?

① Spanning

② LI.

$$aA_1 + bA_2 + cA_3 + dA_4 = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

$$\left. \begin{aligned} a+b+c+d &= 0 \\ b+c+d &= 0 \\ c+d &= 0 \end{aligned} \right\} \Rightarrow \begin{aligned} d &= 0 \\ a &= 0 \\ b &= 0 \\ c &= 0 \end{aligned}$$

Example: Span

Example (Span in $\mathcal{P}_2(\mathbb{R})$)

Does the following set of vectors $\{x^2, x + x^2, 1 + x + x^2\}$ span $\mathcal{P}_2(\mathbb{R})$?

Let's take an arbitrary polynomial

$$p(x) = a_0 + a_1x + a_2x^2$$

$$a_0, a_1, a_2 \in \mathbb{R}$$

$$\begin{aligned} a_0 + a_1x + a_2x^2 &= ax^2 + b(x + x^2) + c(1 + x + x^2) \\ &= c + (b + c)x + (a + b + c)x^2 \end{aligned}$$

$$\left. \begin{aligned} a_0 &= c \\ a_1 &= b + c \\ a_2 &= a + b + c \end{aligned} \right\}$$

Linear Independence of Functions

Definition (Linear Independence)

The set of functions $\{f_1, f_2, \dots, f_k\}$ is linearly independent on an interval I if and only if the only values of scalars $c_1, c_2, \dots, c_k$ such that

$$c_1 f_1(x) + c_2 f_2(x) + \dots + c_k f_k(x) = 0, \quad \forall x \in I,$$

are $c_1 = c_2 = \dots = c_k = 0$.

Remark

This is an important remark that will be displayed in a standard block style. The main point to notice is that the above condition must hold for all x in I .

Definition (Wronskian)

The set of functions $f_1, f_2, \dots, f_k$ be functions in $C^{k-1}(I)$. The Wronskian of these function is the order k determinant defined by

$$W[f_1, f_2, \dots, f_k](x) := \begin{vmatrix} f_1(x) & f_2(x) & \cdots & f_k(x) \\ f_1'(x) & f_2'(x) & \cdots & f_k'(x) \\ \vdots & \vdots & \ddots & \vdots \\ f_1^{(k-1)}(x) & f_2^{(k-1)}(x) & \cdots & f_k^{(k-1)}(x) \end{vmatrix}.$$

Remark

Using the properties of the determinants,

$$W[f_2, f_1, \dots, f_k](x) = -W[f_1, f_2, \dots, f_k](x).$$

Result

Theorem (Linear Independence of Functions)

Let $f_1, f_2, \dots, f_k$ be functions in $C^{k-1}(I)$. If $W[f_1, f_2, \dots, f_k](x_0) \neq 0$ for some $x_0 \in I$, then $\{f_1, f_2, \dots, f_k\}$ is linearly independent on I .

Remark

If $W[f_1, f_2, \dots, f_k](x) = 0$ for all $x \in I$, the above theorem does not give any information as to the linear dependence or independence of $\{f_1, f_2, \dots, f_k\}$.

Example

The set of functions $f_1(x) = x$, $f_2(x) = x + x^2$, $f_3(x) = 2x - x^2$ gives $W[f_1, f_2, f_3](x) = 0$, thus no conclusions can be drawn. However, we can have $f_2 = 3f_1 - f_3$, hence linearly dependent.

$$\begin{vmatrix} f_1 & f_2 & f_3 \\ f_1' & f_2' & f_3' \\ f_1'' & f_2'' & f_3'' \end{vmatrix} = 0 \quad \text{do not conclude.}$$

$$\mathcal{P}_n(\mathbb{R}) \longleftrightarrow \{1, x, x^2, \dots, x^n\} \longrightarrow \text{lin ind.}$$

$$\begin{vmatrix} 1 & x & x^2 & \dots & x^n \\ 0 & 1 & 2x & \dots & nx^{n-1} \\ 0 & 0 & 2 & \dots & n(n-1)x^{n-2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & n! \end{vmatrix} \neq 0$$

$\neq 0$



Row-Echelon Matrices

Definition (Row-Echelon Matrices)

An $m \times n$ matrix is called a row-echelon matrix if it satisfies the following three conditions:

1. If there are any rows consisting entirely of zeros, they are grouped together at the bottom of the matrix.
2. The first non-zero element in any nonzero row is 1 (called a leading 1).
3. The leading 1 of any row below the first row is to the right of the leading 1 of the row above it.

Example

$\begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$ is a row-echelon matrix whereas $\begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 0 & 1 & -1 \end{pmatrix}$ is not.

Reduced Row-Echelon Matrices

(RREF) Form.

Definition

An $m \times n$ matrix is called a reduced row-echelon matrix if it satisfies the following conditions:

1. It is a row-echelon matrix.
2. Any column that contains a leading 1 has zeros everywhere else.

Example (Reduced Row-Echelon Matrices)

$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ is an RREF. Although, $\begin{pmatrix} 2 & \boxed{1} \\ 0 & 1 \end{pmatrix}$ is in REF it's not in RREF.

Thank You

Questions?